

Putting placebo effects in a new light

Placebo effects have puzzled and intrigued researchers for decades but, despite playing an important role in modern trial design and treatment development, they remain relatively mysterious. With the development of new treatments in psychiatry, especially psychedelic-assisted psychotherapy, placebo effects are under the spotlight. In this issue, *The Lancet Psychiatry* presents two Review articles by Matthew Burke and colleagues on reconceptualising placebo and nocebo effects.

Whereas placebo effects are beneficial in clinical care, in research they are often considered more as a nuisance to be rid of—noise that reduces the ability to measure a clear signal. Nocebo effects caused by negative expectations can generate or amplify symptoms or side-effects in clinical settings; while in research, if a participant believes they have not received the desired treatment, nocebo effects can affect measurements of clinical efficacy and promote false-positive findings.

The first paper addresses the neurobiological mechanisms of placebo and nocebo effects—the understanding of which is still at nascent stages—and how these might be harnessed to improve patient outcomes. Burke and colleagues highlight a misperception that hangs over placebo and nocebo responses, adding to their negative connotations: if psychiatric symptoms respond to placebos, then the symptoms must not be real. This false construction not only stigmatises mental illness, but probably prevents the clinical and research communities from exploring the role of placebo effects in clinical care. Burke and colleagues provide several examples of new knowledge of brain circuitry and placebo responses, demonstrating that some of the very circuits targeted by new treatments are also activated by placebos. Rather than discounting placebo effects, researchers should aim to harness them to improve outcomes through synergising placebo responses with responses to the active ingredients of treatments. Doing so, however, is technically challenging. Similarly, the ethical issues surrounding placebos and informed consent for patients are vigorously debated. Burke and colleagues provide several recommendations for where the field can improve the use of placebos in clinical care.

The second paper focuses on placebo effects in clinical trials, particularly for newer interventions

such as psychedelics and device-based interventions (eg, transcranial magnetic stimulation).

Some sham-controlled trials of surgical and device-based interventions have demonstrated relatively high placebo responses, with one double-blind sham-controlled study reporting a 41% response to sham treatment. The authors discuss several potential reasons for such high rates of participants responding to sham treatment, compared with other treatment modalities (eg, oral drug treatments), including excitement about a perceived innovation, attention paid to participants receiving both intervention and sham being greater than in standard clinical care, and increasing personalisation of device-based treatments, all of which can influence expectations and increase placebo responses. For psychedelic-assisted psychotherapy, the authors describe the specific challenges researchers face with masking. Several high-profile studies demonstrating efficacy for psychedelics such as psilocybin have either not recorded whether participants were successfully masked to treatment allocation or, if masking was tested, have shown that the majority of participants knew to which group they were assigned. This has major implications for trying to demonstrate clear efficacy beyond the placebo response, and because of the relatively obvious and quick responses people have to psychedelics, solving this problem is not trivial. Burke and colleagues discuss recent attempts to improve masking in psychedelic research, such as using varying doses, and provide specific recommendations to improve the reporting and design of trials to better understand the role of masking and placebo responses.

A clear message from these papers is that regardless of how one views placebo and nocebo effects—either as friend or foe—they are pervasive and growing in importance as our treatment modalities change. Rather than wishing them away in narrow attempts to demonstrate a treatment's efficacy, researchers should try to better understand placebo and nocebo responses to help control them and to harness their therapeutic potential alongside active treatments. Given the unique importance of placebo and nocebo responses in mental health, psychiatry should lead the way. The papers by Burke and colleagues shed light on the direction that should be taken. ■ *The Lancet Psychiatry*



For more on placebo and nocebo effects see [Review](#) pages 413 and 426

For more on sham-controlled device-based treatments see [Articles](#) *Lancet Psychiatry* 2023; 10: 172–83